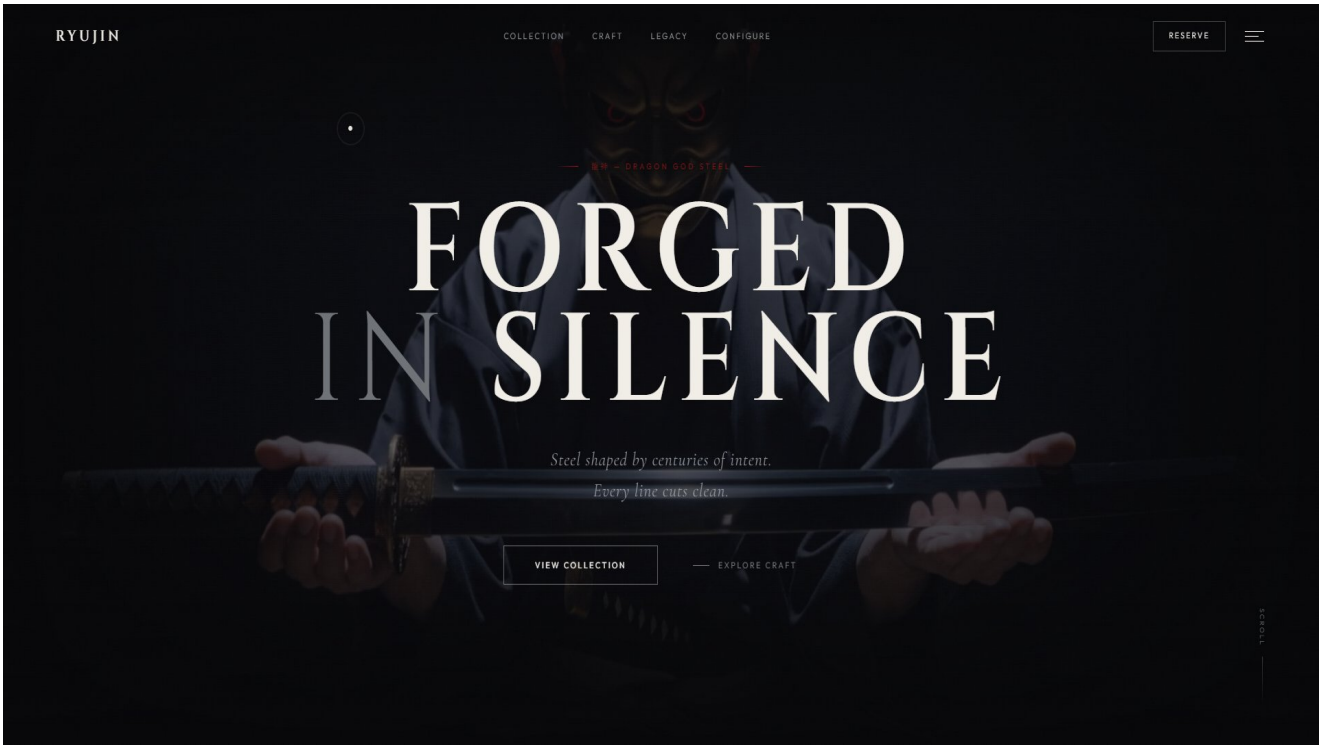


CASE STUDY

RYUJIN

Luxury Katana E-Commerce & Configurator

ROLE	TYPE	STACK	YEAR
UI/UX Design Front-End Dev	Personal Project Portfolio Piece	HTML · CSS · JS Three.js · GSAP	2025



Hero section — RYUJIN: Forged in Silence

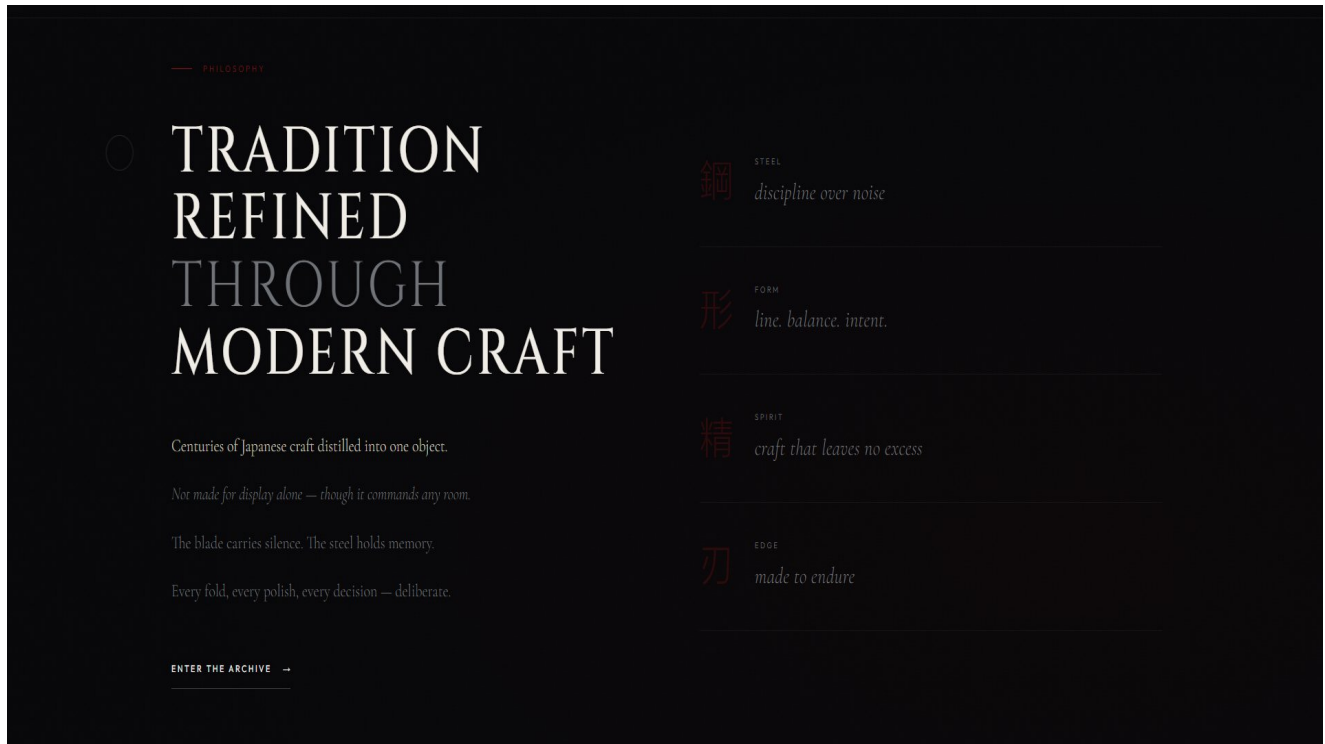
OVERVIEW

About the Project

RYUJIN is a concept luxury katana brand and e-commerce experience built entirely as a single-page application. The project explores how high-end physical craftsmanship can be communicated through cinematic web design — treating the browser as a gallery, not a catalogue.

The challenge was to build something that felt as intentional and refined as the product it represents. Every section of the site has a distinct purpose: seduce, educate, personalise, convert.

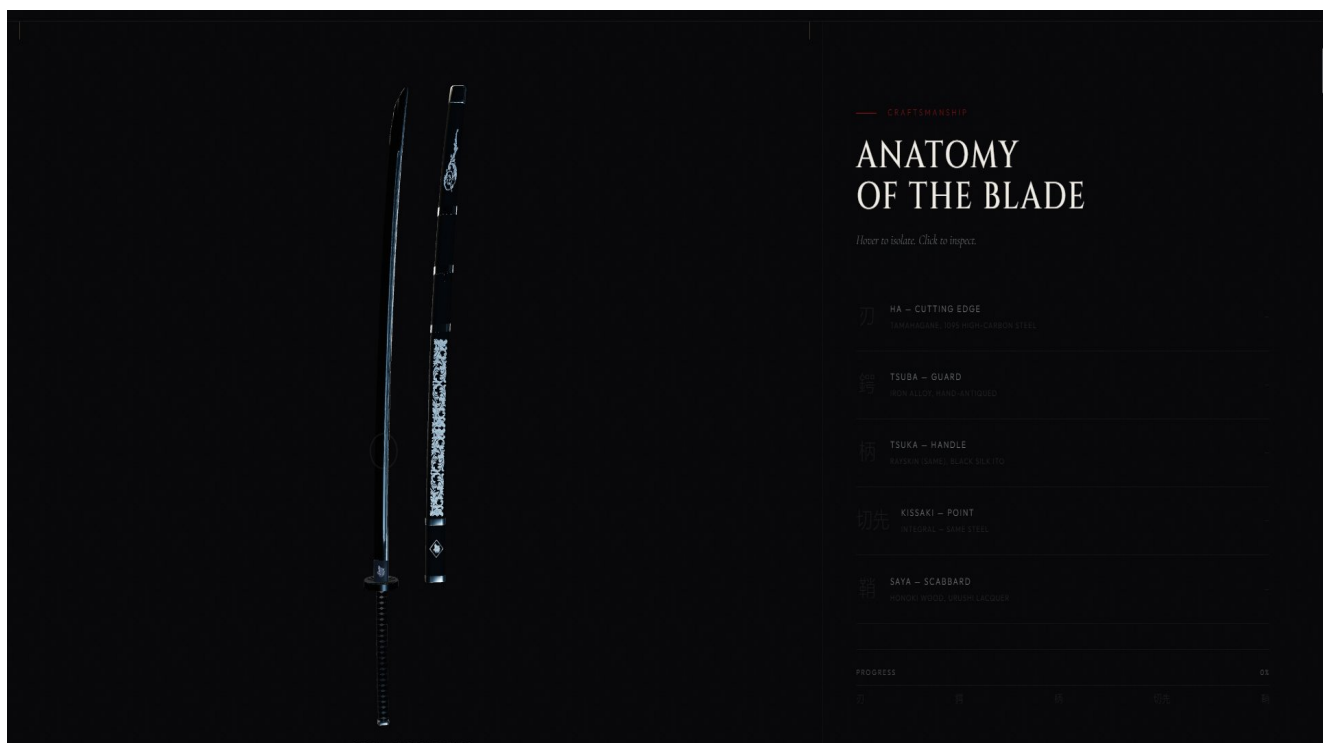
DESIGN DIRECTION



Philosophy section — Tradition Refined Through Modern Craft

The visual language draws from Japanese aesthetics: restraint, negative space, and precision. The dark palette, serif typography, and red accent mirror the weight of the product — nothing is decorative that isn't also functional.

INTERACTIVE 3D — ANATOMY OF THE BLADE



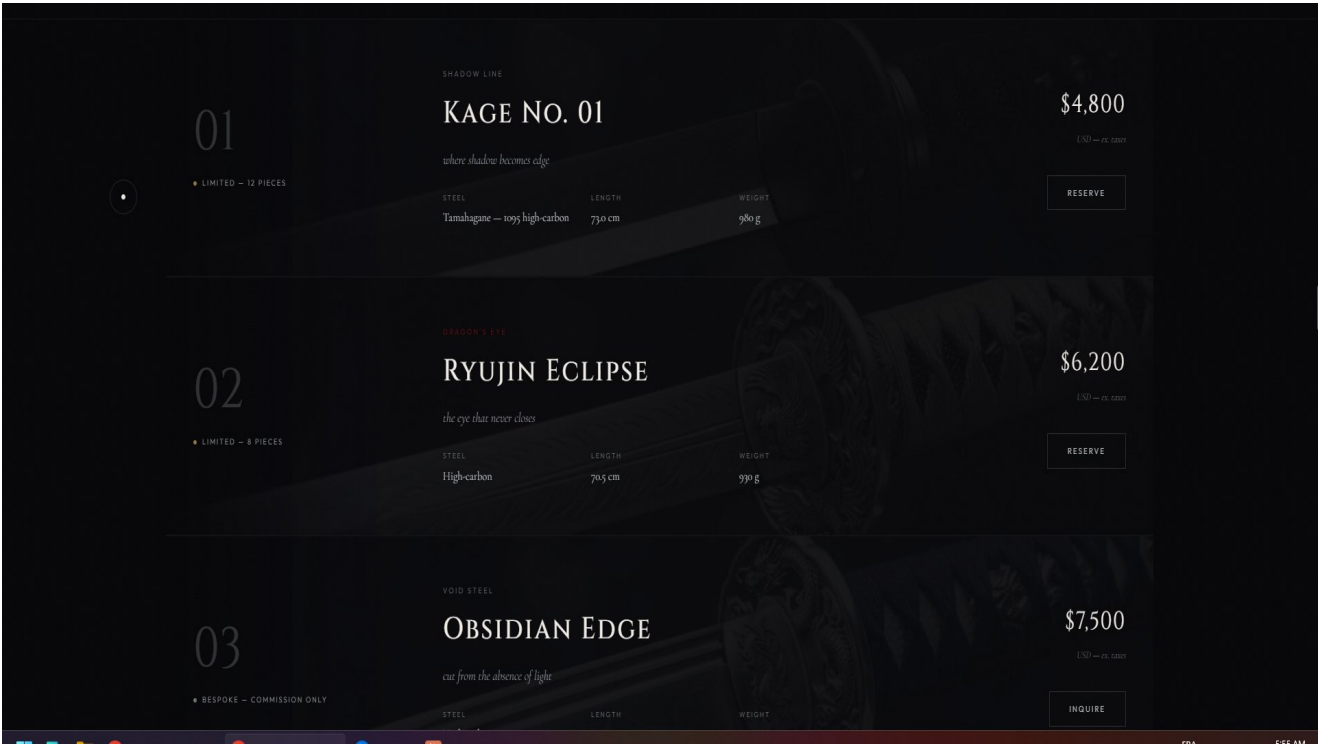
Anatomy section — hover to isolate, click to inspect each blade component

One of the core interactions is the blade anatomy viewer. Built with Three.js, the katana model is broken into seven named mesh parts — Ha (cutting edge), Tsuba (guard), Tsuka (handle), Kissaki (point), Saya (scabbard), and more. Users can hover to isolate individual components and click to inspect material specifications.

- 3D model exported as GLB from Blender with seven distinct named mesh parts
- Per-mesh material swapping driven by user interaction

- Smooth isolation animations using Three.js and GSAP timelines
- Progress tracker showing scroll depth through the anatomy section

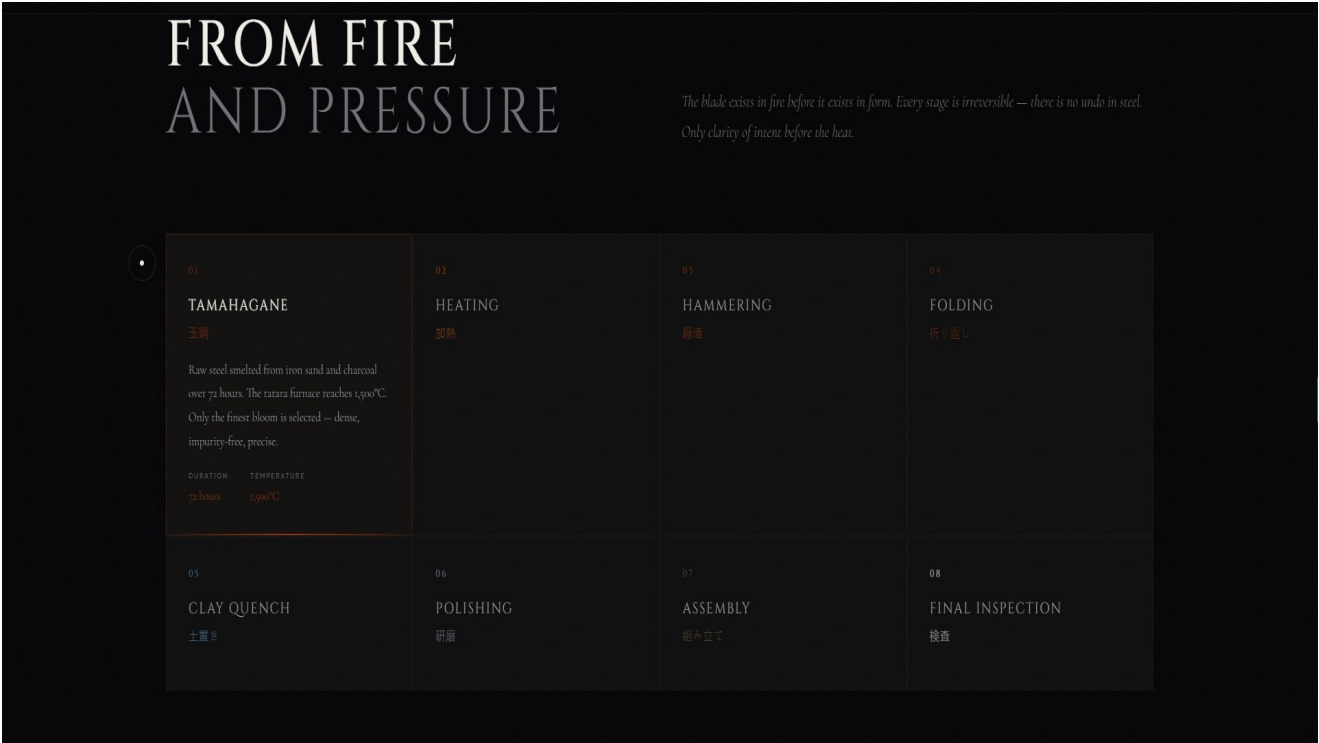
COLLECTION & PRODUCT LISTING



Collection section — limited series with pricing, specs, and reservation flow

The collection page presents three katana series — Kage No. 01, Ryujin Eclipse, and Obsidian Edge — as editorial entries rather than product cards. Steel type, blade length, weight, edition size, and price are surfaced cleanly without cluttering the experience.

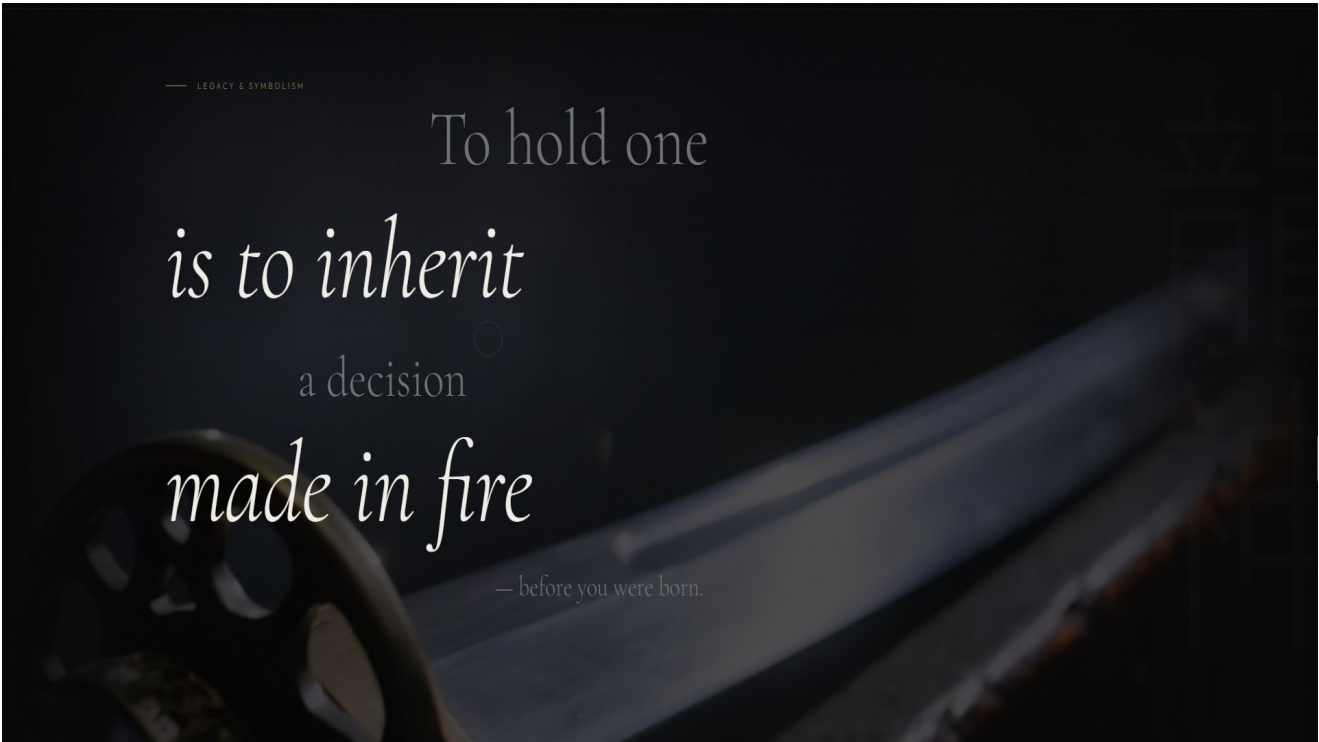
CRAFT PROCESS — FROM FIRE AND PRESSURE



Craft section — 8-stage forging process with interactive cards

An eight-stage interactive grid documents the traditional Japanese forging process from Tamahagane smelting to final inspection. Each card reveals duration, temperature, and a description of the stage — educating the collector while deepening the brand story.

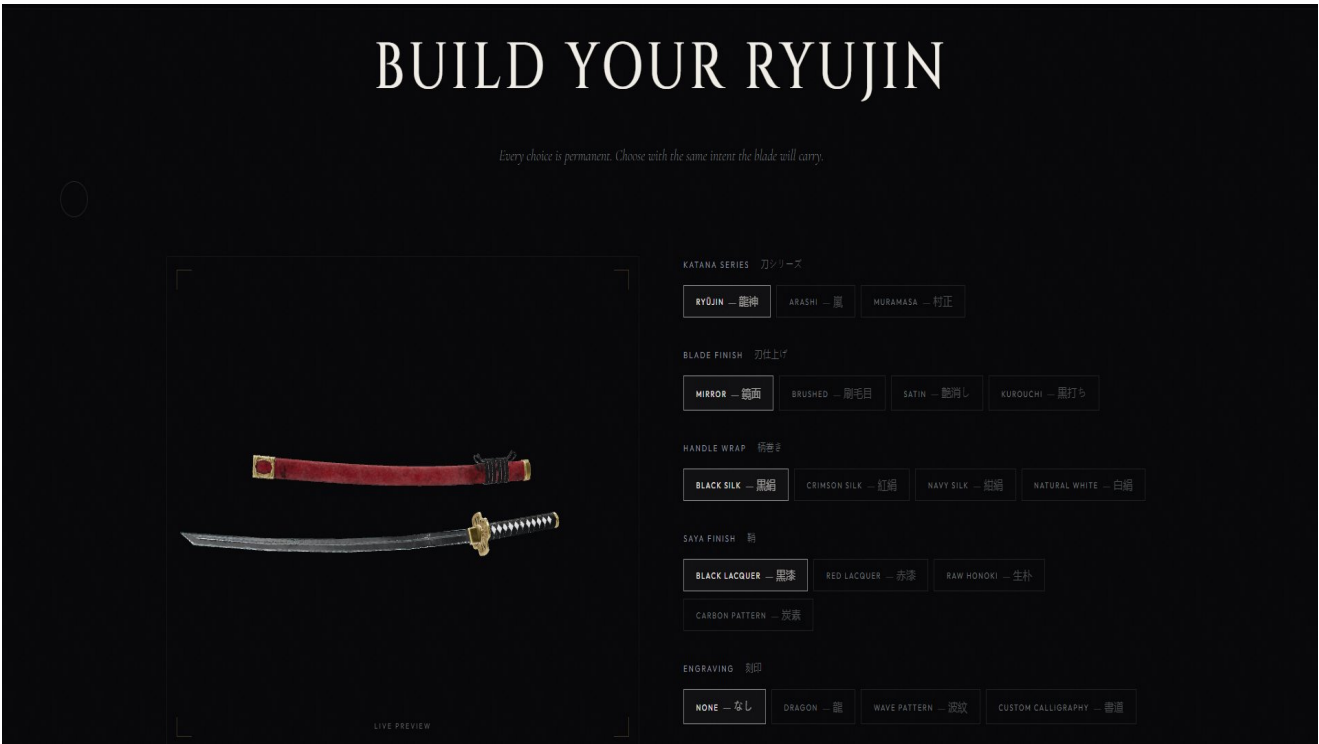
LEGACY & STORYTELLING



Legacy section — kinetic typography driven by scroll

The legacy section uses scroll-driven kinetic typography — large type animating at different speeds as the user moves through the page. No UI chrome, no buttons. Pure atmosphere built with GSAP ScrollTrigger.

LIVE CONFIGURATOR — BUILD YOUR RYUJIN

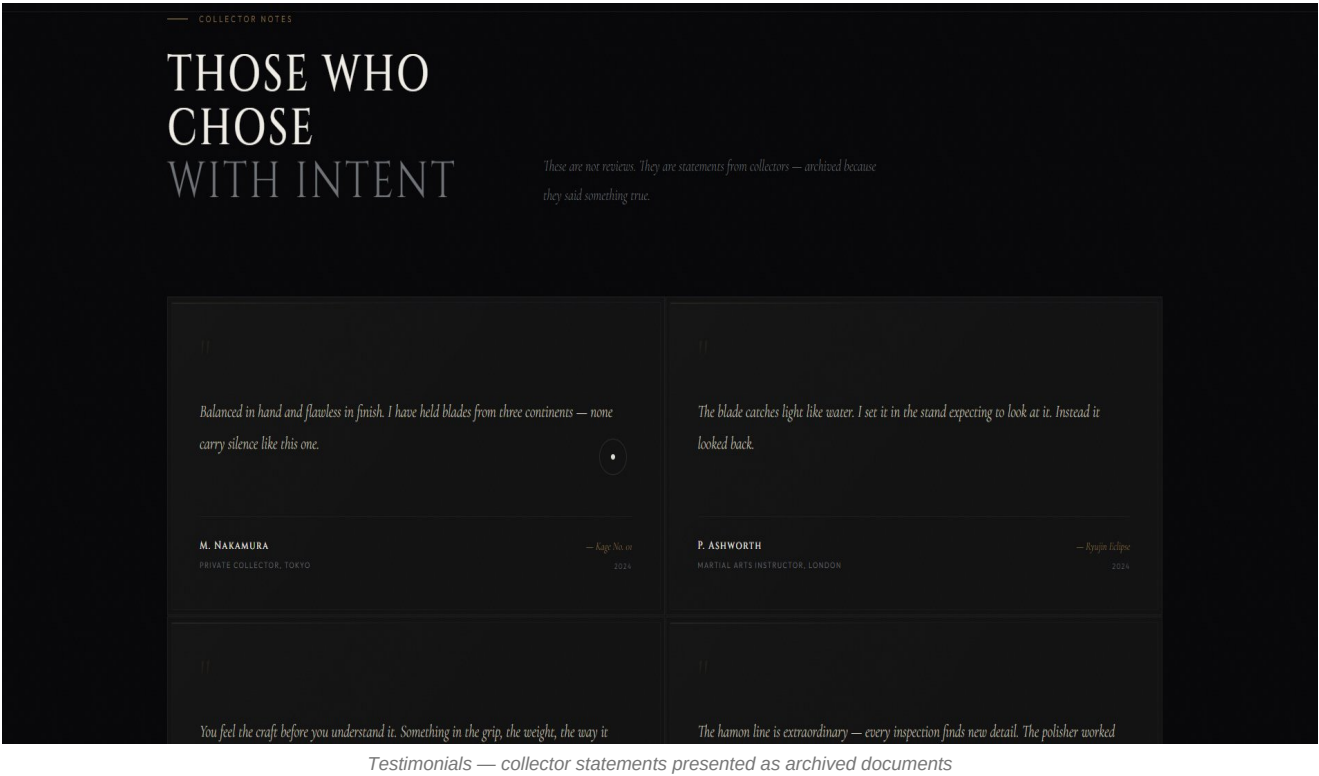


Configurator — real-time katana customisation with live 3D preview

The configurator is the centrepiece of the experience. Users select from multiple options across five dimensions — Katana Series, Blade Finish, Handle Wrap, Saya Finish, and Engraving — with the 3D model updating in real time to reflect each choice.

- Live material swapping on the Three.js GLB model — no page reload
- Four blade finishes: Mirror, Brushed, Satin, Kurouchi
- Four handle wrap options in Japanese and English labelling
- Five Saya finishes including Carbon Pattern
- Custom calligraphy engraving option

COLLECTOR NOTES



Rather than conventional reviews, the social proof section is framed as archived collector statements — a deliberate editorial choice that maintains the brand's tone of exclusivity and restraint.

NOT EVERY BLADE BELONGS TO EVERY HAND



Closing section — scarcity messaging with series and edition count

TECHNICAL STACK

3D & Animation	Three.js (GLB model, mesh isolation, real-time material swapping), GSAP ScrollTrigger
Frontend	Vanilla HTML, CSS, JavaScript — no framework, fully hand-crafted
3D Modelling	Blender — katana modelled with seven named mesh parts, exported as GLB
Design	Custom type system, dark luxury palette, Japanese kanji integrated into UI
Deployment	GitHub Pages — htom666.github.io/ryujin.github.io

Live site: htom666.github.io/ryujin.github.io · GitHub: github.com/htom666